

1.(18%) Please calculate 149 divided by 26 using the hardware described in Figure 5. (Assume 149 and 26 are decimal numbers). You should show the contents of each register on each step. Assume A and B are unsigned 8-bit integers. You will want to think hard about this, do an experiment or two, or else go to the web to figure out how to make this work correctly. (Hint: one possible solution involves using the fact that Figure 5. implies the remainder register can be shifted either direction.)

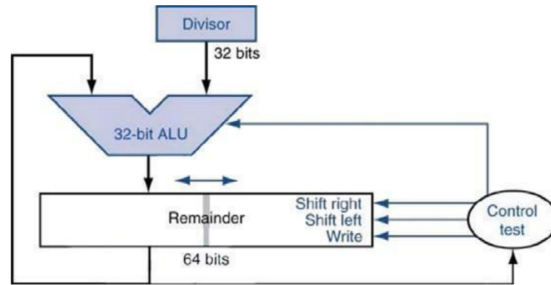


Figure 5. Division hardware

Start : Place dividend in remainder

1 : Shift remainder register left 1 bit

2 : Subtract divisor register from the left half of remainder register,
and place the result in the left half of remainder register.

3a : Shift remainder to left, set new rightmost bit to 1

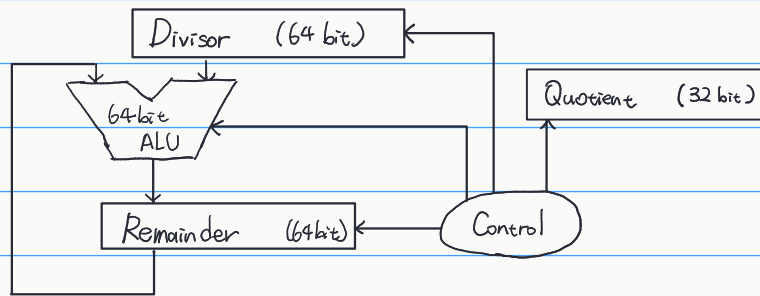
3b : Restore original value by adding divisor to the left half of remainder, and place sum in the left half of remainder. Also shift remainder to left, setting the new right most bit to 0.

Done : Shift left half remainder right 1

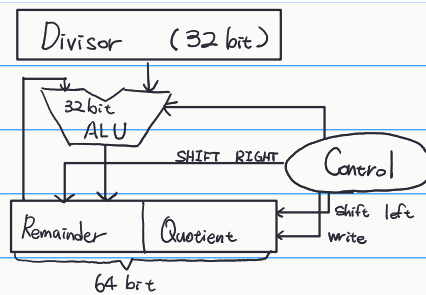
Iteration	Step	Remainder(R)	Divisor
0	Start 1	0000 0000 1001 0101 0000 0001 0010 1010	0001 1010
1	2 3b	1110 0111 0010 1010 0000 0010 0101 0100	0001 1010
2	2 3b	1110 1000 0101 0100 0000 0100 1010 1000	0001 1010
3	2 3b	1110 1010 1010 1000 0000 1001 0101 0000	0001 1010
4	2 3b	1111 0111 0101 0000 0001 0010 1010 0000	0001 1010
5	2 3b	1111 1000 1010 0000 0010 0101 0100 0000	0001 1010
6	2 3a	0000 1011 0100 0000 0001 0110 1000 0001	0001 1010
7	2 3b	1111 1010 1000 0001 0010 1100 0000 0010	0001 1010
8	2 3a	0001 0011 0000 0010 0010 0110 0000 0101	0001 1010
9	Done	0001 0011 0000 0101	0001 1010

2.(20%) Given the basic division algorithm hardware architecture,

(a) (8%) Draw the hardware architecture diagram of the Version 1 Divider.



(b) (7%) Draw the hardware architecture diagram of the Version 2 Divider.



(c) (5%) Compare the two versions of dividers and describe the main advantages of Version 2.

A: 第二版的除法器可以節約暫存器使用量

3.(40%) Given the IEEE-754 single precision floating - point operand format as follows,

Sign (s)	Exponent (e)	Fraction (f)
1 bit	8 bits	23 bits

(a)(6%) What is the decimal number -39.25 represented in binary number?

$$39.25_{(10)} = 100111.01_{(2)} = 1.0011101 \times 2^5$$

$$5 + 127 = 132$$

$$A: \begin{array}{c|c|c} \text{S} & \text{Exponent} & \text{Fraction} \\ \hline 1 & 1000100 & 0011010000000000000 \end{array}$$

(b)(6%) If sign=1, exponent = 01111000, and significand = 01001000000000000000000.
What is the decimal number?

$$01111000_{(2)} = 120_{(10)} \quad 120 - 127 = -7$$

$$1.01001_{(2)} \times 2^{-7} = \frac{41}{32} \times \frac{1}{2^7} = 0.0100098 \quad A: -0.0100098$$

(c)(5%) What is the largest positive normalized number in IEEE-754?

$$\begin{array}{c|c|c} 0 & 11111111 & 11111111111111111111111 \\ \hline \text{S} & \text{Exponent} & \text{Fraction} \end{array} = 1.11111111111111111111111 \times 2^{127}$$

$$\approx 2 \times 2^{127} \approx 3.4 \times 10^{38} \quad A: 3.4 \times 10^{38}$$

(d)(5%) What is the smallest positive denormalized number in IEEE-754?

$$\begin{array}{c|c} 0 & 00000000 \\ \hline \text{S} & \text{Exponent} \end{array} \quad \begin{array}{c|c} 00000000 & 000000000000000001 \\ \hline \text{Fraction} & \end{array}$$

$$2^{-23} \times 2^{-126} = 2^{-149} \approx 1.4 \times 10^{-45} \quad A: 1.4 \times 10^{-45}$$

(e)(12%) What are the $\pm\infty$ and NaN in IEEE-754 Single Precision?

Sign	Exponent	Fraction	Object represented
0	255	0	$+\infty$
1	255	0	$-\infty$
0/1	255	Non-Zero	NaN

(f)(6%) Convert the IEEE 754 single precision floating-point number 0xC102B132, represented in hexadecimal, to its equivalent decimal number.

$$0 \times C102B132_{(16)} = \begin{array}{c|c|c|c|c|c|c|c} 1 & 100 & 0001 & 0000 & 0010 & 1011 & 0001 & 0011 & 0010 \\ \hline & & & & & & & & 130 \end{array}$$

$$130 - 127 = 3$$

$$1.0000010101100010011001 \times 2^3 \approx 8.1682 \quad A: -8.1682$$

4.(22%) IEEE 754-2008 contains a half precision that is only 16 bits wide. The left most bit is still the sign bit, the exponent is 5 bits wide and has a bias of 15, and the mantissa is 10 bits long. A hidden 1 is assumed.

(a)(6%) Write down the bit pattern to represent -1.4375 assuming a version of IEEE 754-2008 format.

$$1.4375_{(10)} = 1.0111_{(2)} \times 2^0$$

$$\text{Ans: } \begin{array}{c} \text{S} \quad \text{EXPONENT (5)} \quad \text{MANTISSA (10)} \\ 1 \quad 0111 \quad 0111000000 \end{array}$$

(b)(8%) Calculate the sum of 1.84375×10^2 and $3.642578125 \times 10^{-1}$ by hand. Assume 1 guard, 1 round bit, and 1 sticky bit, and round to the nearest even. Show all the steps.

$$\textcircled{1} \quad 1.84375 \times 10^2 = 184.375 = 10111000.011_{(2)} = 1.0111000011 \times 2^7$$

$$\textcircled{2} \quad 3.642578125 \times 10^{-1} = 0.3642578125 = 0.010110101_{(2)} = 1.0110101 \times 2^{-2} = 0.00000000101101 \times 2^7$$

$$\textcircled{1} \quad 1.0111000011$$

$$+ \textcircled{2} \quad 0.00000000101 \dots$$

$$\begin{array}{r} 1.011100010001 \\ \text{LSB GR} \end{array}$$

GR = 01 $\rightarrow$ 捨去

$$\hookrightarrow 1.0111000100 \times 2^7$$

$$A: 010110011000100$$

(c)(8%) Calculate the product of 1.4150390625 and 8.59375 by hand. Assume 1 guard, 1 round bit, and 1 sticky bit, and round to the nearest even. Show all the steps.

$$1.4150390625_{(10)} = 1.0110101001_{(2)} = 1.0110101001 \times 2^0$$

$$8.59375 = 1000.10011_{(2)} = 1.00010011 \times 2^3 =$$

$$1.0110101001$$

$$\times 1.00010011$$

$$\begin{array}{r} 1111212 \\ 10110101001 \end{array}$$

$$10110101001$$

$$10110101001$$

$$10110101001$$

$$\begin{array}{r} 11000101010010011 \times 2^3 \\ \text{LSB GR S} \end{array}$$

$$\text{MSB} = 0 \quad \text{GR} = 10 \quad \text{S} = 1 \rightarrow \lambda \text{ (進位)}$$

$$\hookrightarrow 1.100010101 \times 2^3$$

$$A: 010010100010101$$